Wireshark Traffic Analysis Lab

Overview

This lab demonstrates capturing and analyzing network traffic using Wireshark.

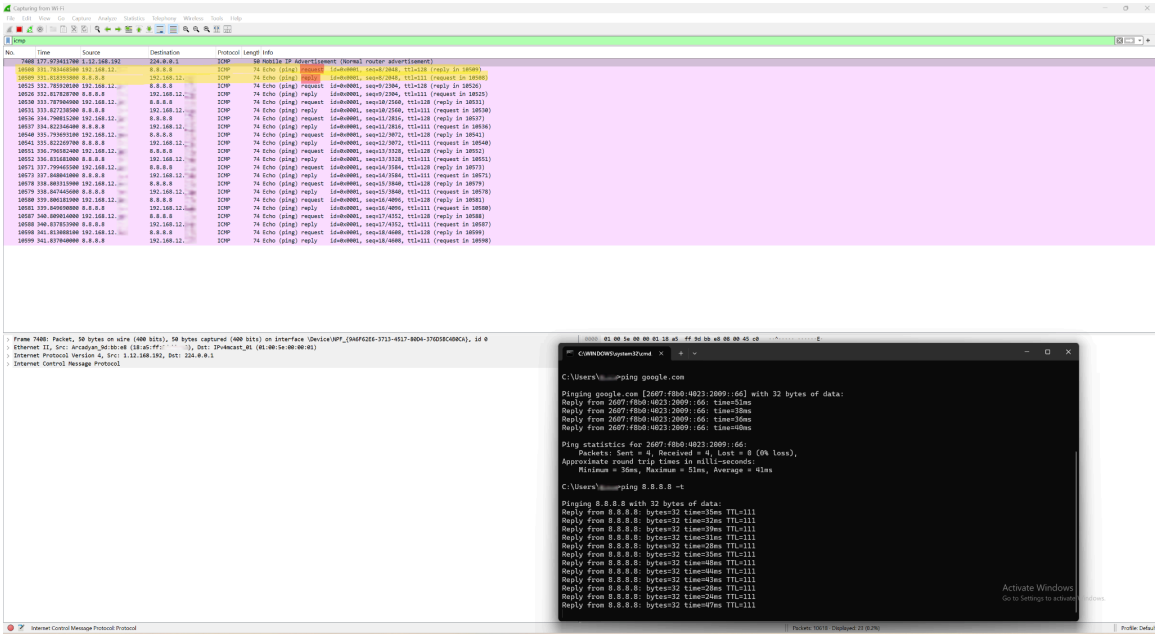
Objectives

- Capture live network traffic
- Analyze ICMP, DNS, and TCP/TLS traffic
- Apply filters to isolate protocols

Environment

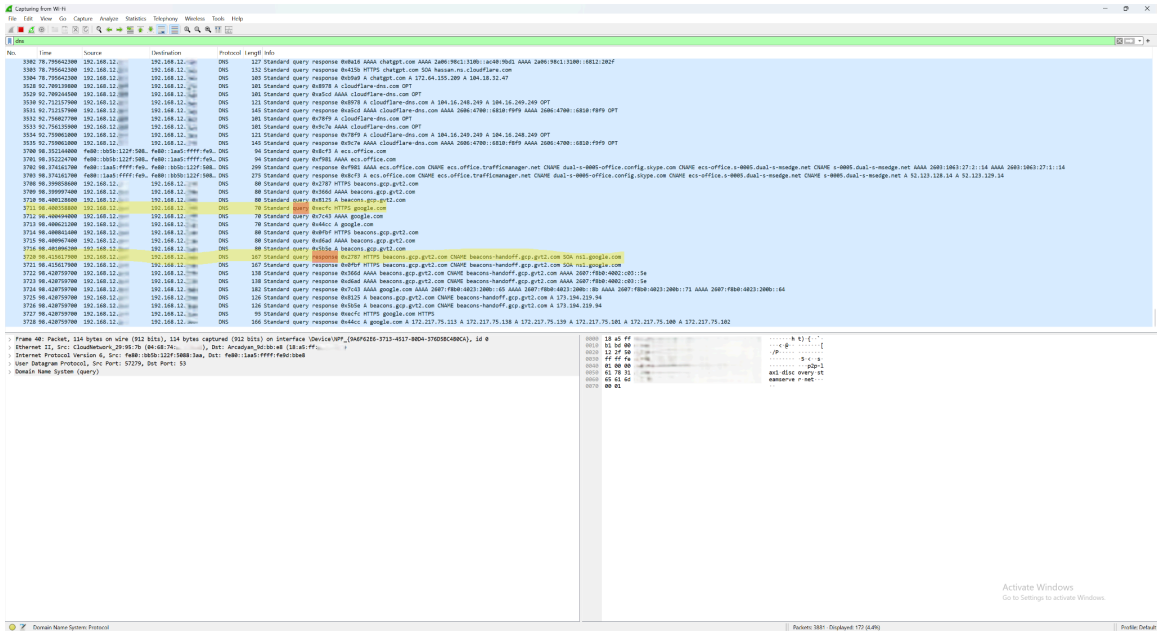
- Windows 11
- Wireshark

Part 1: ICMP Traffic Analysis



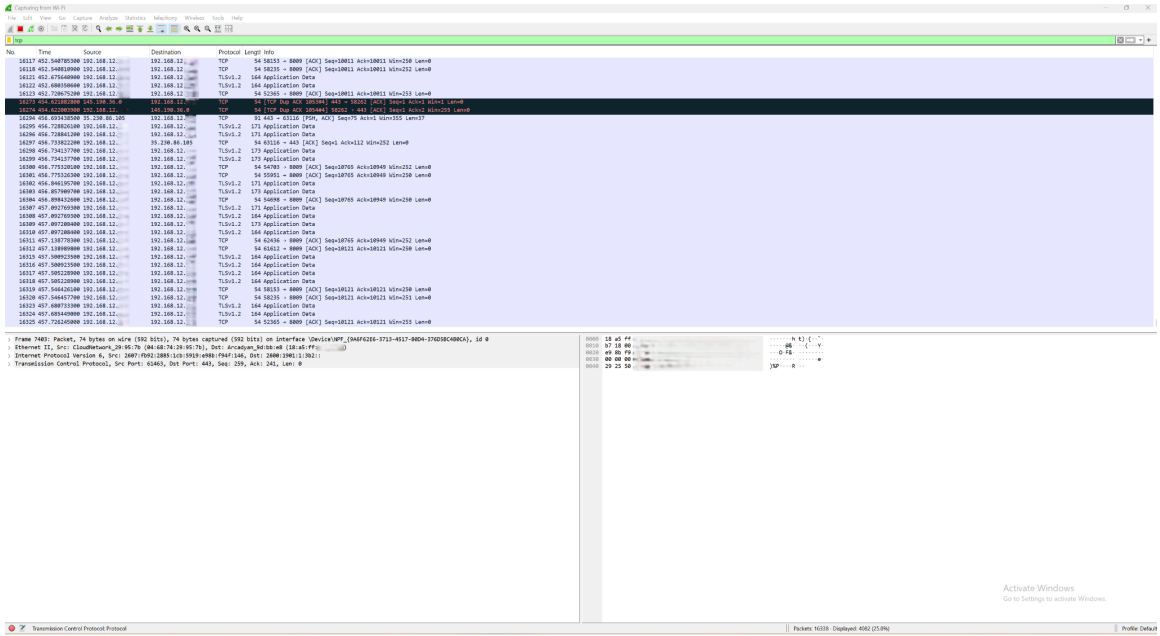
Description:
ICMP traffic was analyzed by generating ping requests from the client machine to an external host (8.8.8.8). Using the ICMP filter in Wireshark, echo request and echo reply packets were identified. This confirms successful network connectivity and demonstrates how ICMP is used for basic network diagnostics.

Part 2: DNS Traffic Analysis



Description:
DNS traffic was captured and analyzed by applying a DNS filter in Wireshark. The client generated queries for domain names such as google.com, and corresponding responses from the DNS server returned the resolved IP addresses. This demonstrates how domain name resolution occurs in a network environment.

Part 3: TCP/TLS Traffic Analysis



Description:
TCP and TLS traffic was captured while accessing web services. Analysis of the packets shows TCP communication and encrypted TLS sessions between the client and external servers. This demonstrates how application data is securely transmitted over the network.

Findings

- ICMP confirmed connectivity
- DNS resolved domain names
- TCP/TLS showed application traffic

Conclusion

This lab demonstrated real-world packet analysis and troubleshooting using Wireshark.

Notes

Internal IPs and MAC addresses were masked for privacy.